

What is a matrix?

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

2 rows x 3 columns

What are they for?

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Data: Grades? Pixels?

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Transformations

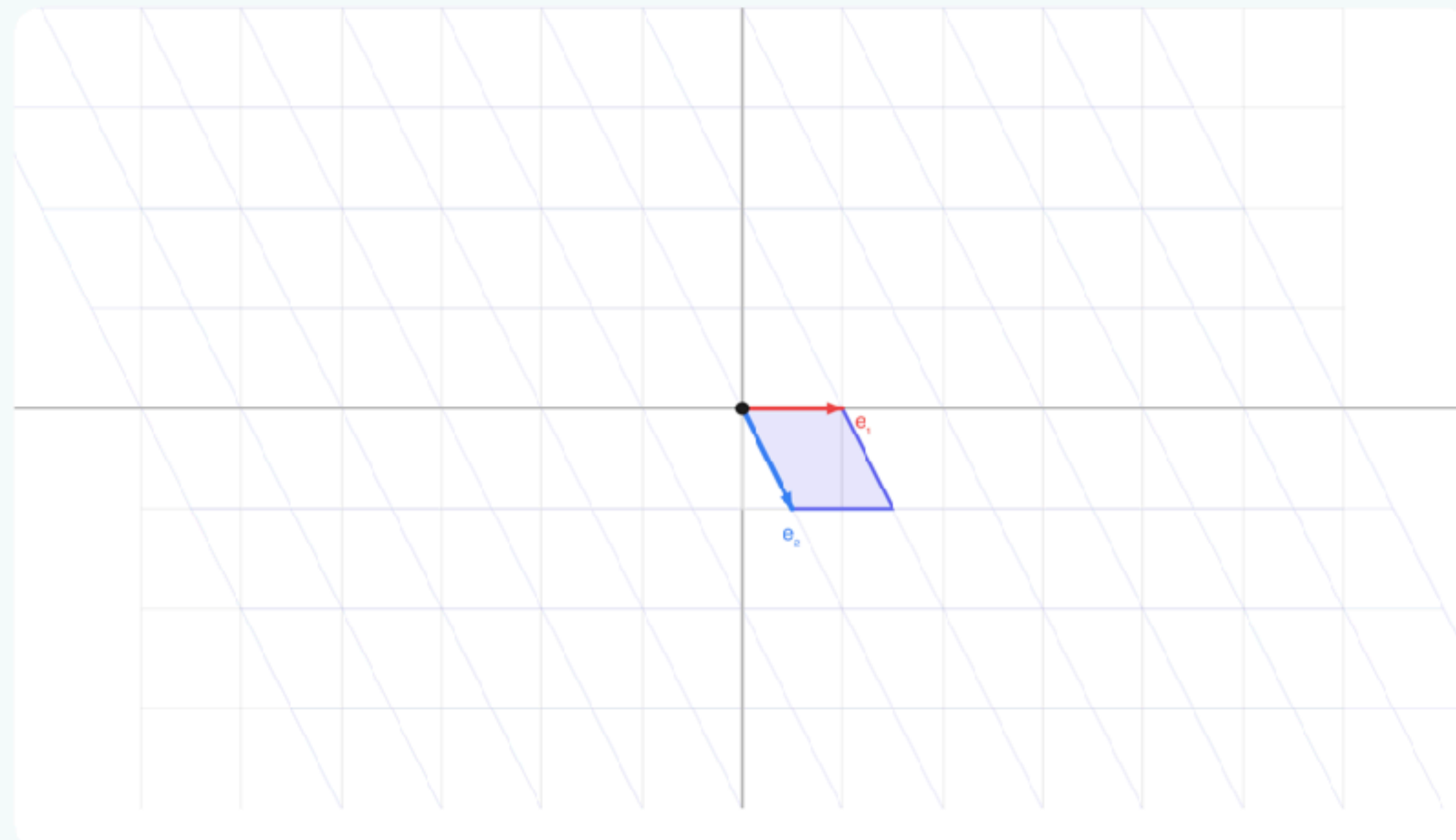
Transformation matrix = function

$$\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x + y \\ 3y \end{bmatrix}$$

Geometric interpretation

Matrix Transformer

Enter a 2×2 matrix or pick a preset. Watch the unit square transform with animated basis vectors and a warped grid.



MATRIX

A =

1.00	0.50
0.00	1.00

PRESETS

Rotate 45°	Reflect Y
Shear X	Scale 2×
Swap XY	Reflect X
Scale ½	Identity

PROPERTIES

det(A) 1.0000
Preserves orientation



<https://8gwifi.org/exams/visual-math/matrix-transform.jsp>

How about non-square matrices?

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

Matrix multiplication against vector

$$\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x + y \\ 3y \end{bmatrix}$$

Computes new vector coordinates

Matrix multiplication against matrix

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \quad R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

RS means apply S then R

$$RS = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 0 & -3 \\ 2 & 0 \end{bmatrix}$$

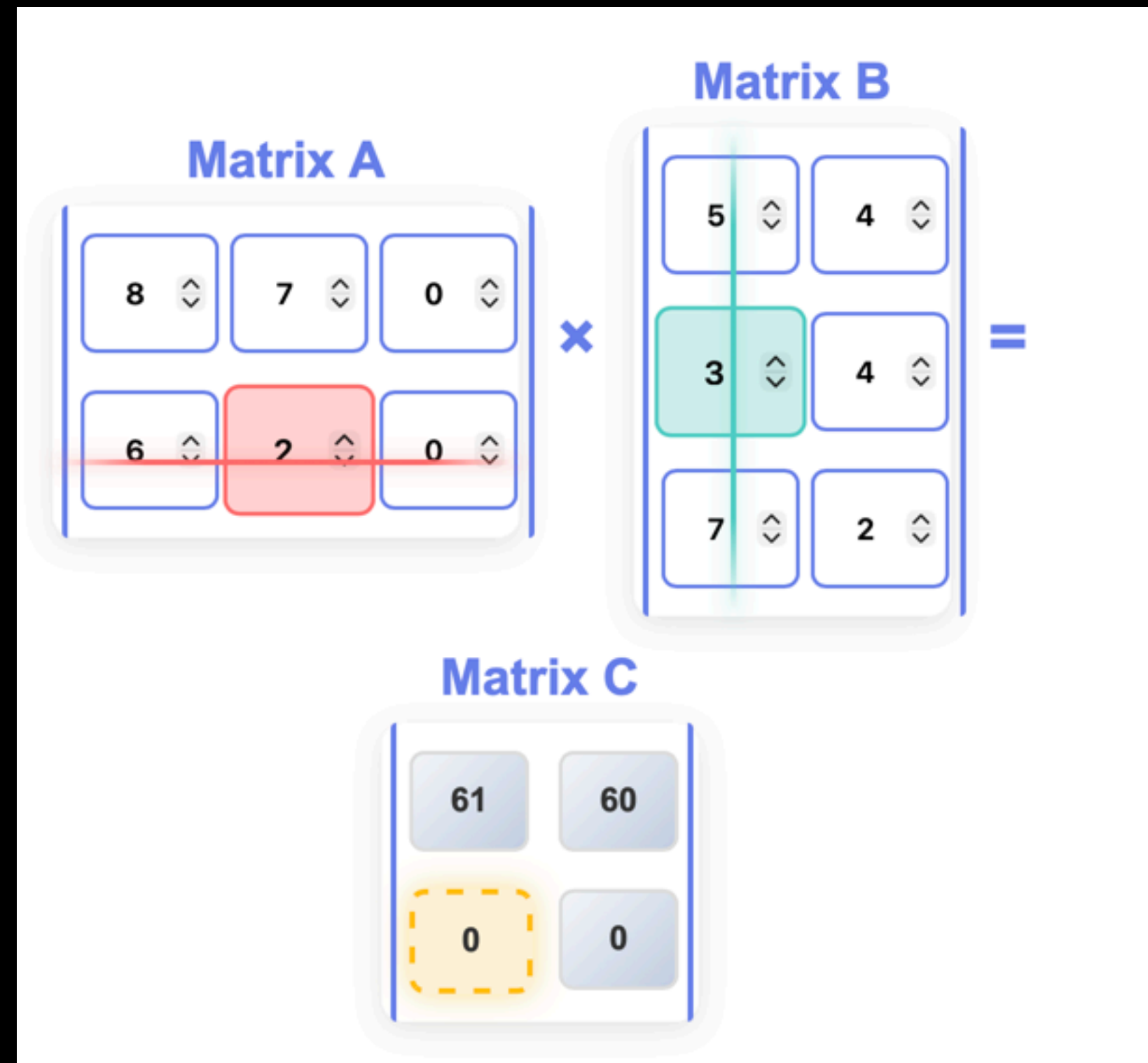
Matrix multiplication order

SR means apply R then S

$$SR = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ 3 & 0 \end{bmatrix}$$

Order matters!

How to multiply?



<https://www.basic-mathematics.com/matrix-multiplication-visualizer.html>

Matrix Multiplication: Output Size

$$(\mathbf{m} \times \mathbf{n}) (\mathbf{n} \times \mathbf{p}) = (\mathbf{m} \times \mathbf{p})$$

First matrix contributes
the number of **rows** (**m**)

Inner dimensions
must match

Second matrix contributes
the number of **columns** (**p**)

The inner dimensions must match. The output uses the outer dimensions.

Example:

$$(\mathbf{2} \times \mathbf{3}) (\mathbf{3} \times \mathbf{4}) = (\mathbf{2} \times \mathbf{4})$$

2 rows

4 columns



Note:

You can multiply only if
the inside numbers are equal.

System of Equations

$$2x + y = 8$$

$$x - y = 1$$

$$\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 1 \end{bmatrix}$$

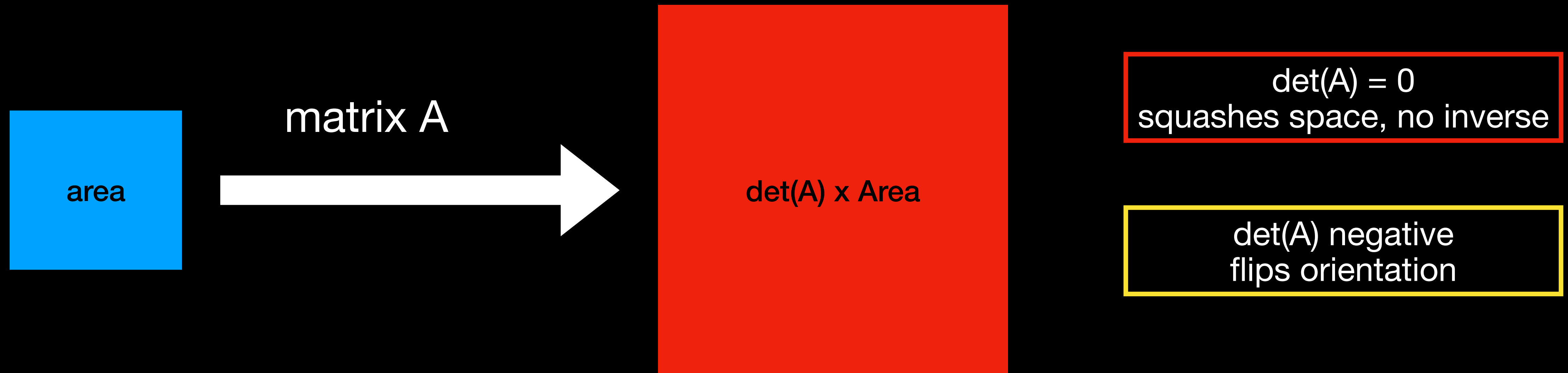
Solving the equation

Many many ways

- Graphical method
- Substitution
- Gaussian elimination
- Gauss–Jordan elimination
- Inverse matrix method
- Cramer's rule
- LU decomposition
- QR decomposition
- Cholesky decomposition
- Iterative methods, such as Jacobi or Gauss–Seidel
- Least-squares method for overdetermined systems
- Pseudoinverse method for non-square or singular systems

Determinant

Measures how a matrix scales the area/volume



$\det(A)=6$ means area becomes 6x larger

Determinant of 2x2

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Determinant of 3x3

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$

Solve via Cramer's Rule

<https://matrix.reshish.com/cramer/>

Identity and Inverse Matrix

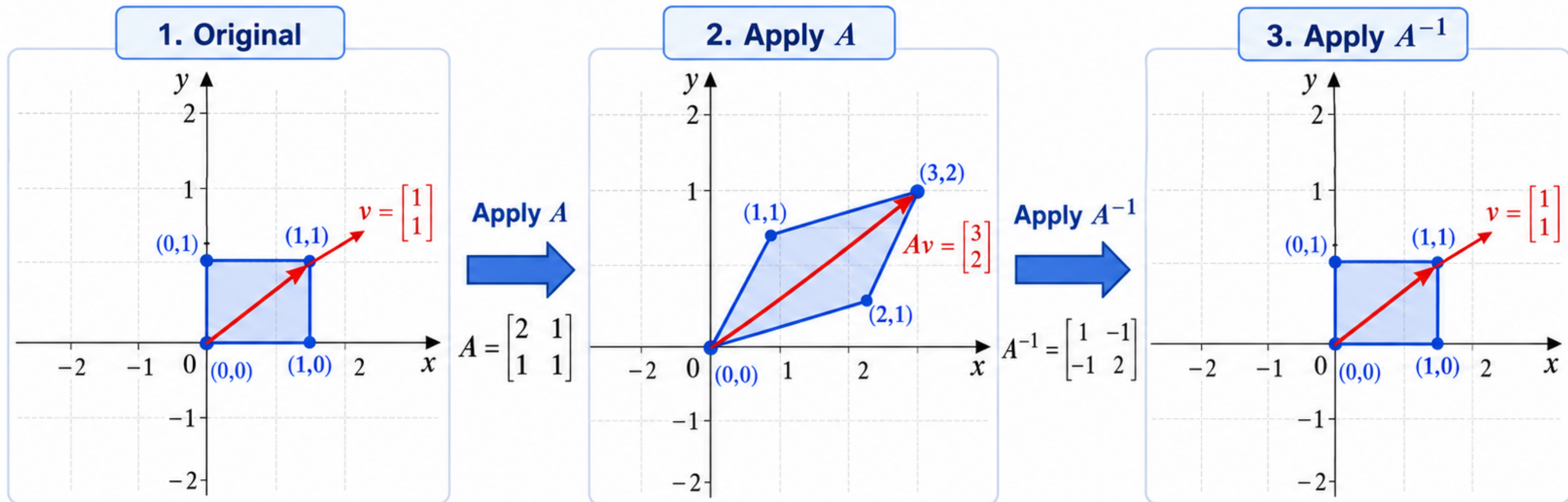
$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^{-1} \cdot A = I$$

<https://matrix.reshish.com/inverse-matrix/>

Inverse Matrix: Undoing a Transformation

A^{-1} reverses the effect of A



The inverse matrix **undoes** a transformation.

If you transform a shape or point with A , then applying A^{-1} brings it back.

i Only possible when $\det(A) \neq 0$

Gauss Jordan Elimination

3 Possible Row Operations

$$R_1 \leftrightarrow R_2$$

$$R_1 \rightarrow cR_1$$

$$R_1 \rightarrow R_1 + cR_2$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{4}{3} \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & -2 \\ 1 & 2 \end{bmatrix}$$

$$R_1 \rightarrow \frac{1}{3}R_1$$

$$R_1 \rightarrow R_1 - 3R_2$$